BEFORE THE NEW YORK STATE
PUBLIC SERVICE COMMISSION

Proceeding on Motion of the Commission as to the Rates, Charges, Rules and Regulations of The Brooklyn Union Gas Company d/b/a National Grid NY for Gas Service. **Case 19-G-0309**

Proceeding on Motion of the Commission as to the Rates, Charges, Rules and Regulations Of The Brooklyn Union Gas Company d/b/a National Grid NY for Gas Service. **Case 19-G-0310**

SUPPLEMENTAL TESTIMONY OF KIM FRACZEK AND LEE ZIESCHE ON BEHALF OF SANE ENERGY PROJECT

January 29, 2020

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I. INTRODUCTIONS AND BACKGROUND

Q.Please state your name and business address.

A. Kim Fraczek and Lee Ziesche. 140 2nd Avenue, New York, NY 10003

Q. What is your occupation?

A:(FRACZEK) Director of Sane Energy Project
(ZIESCHE) Community Engagement Coordinator of Sane Energy Project

Q.Have you previously testified before the Commission?

A:Yes. We recently provided direct testimony for the Con Edison rate case. We have also provided written comments and attended multiple PSC public hearings on numerous PSC cases and have helped New Yorkers understand the public commenting processes and submit thousands of comments on numerous cases before the PSC on energy issues.

Q. What is Sane Energy Project

A. Sane Energy Project is a non-profit organization representing 8,500 New Yorkers. We are based in New York City, and work with communities across New York State to stop the buildout of fracked gas infrastructure and hasten the just transition to renewable energy. Since 2011, we have worked directly with scientists, health professionals, and impacted communities across the state to raise awareness about the health and climate impacts of fracked gas infrastructure, and we work with communities to identify and remove barriers to community-owned renewable energy. We've worked with thousands of New Yorkers to prepare and provide public comment and participate in the regulatory process for over a dozen fracked gas infrastructure projects across New York.

Q.Please summarize your background.

A:(FRACZEK) I have been an employee of Sane Energy Project since 2014. Prior to this, was a volunteer coordinator for Sane Energy Project since its inception in 2011. My background is 12 years in corporate creative direction and digital rapid prototyping leading teams of designers in New York, USA, Tokyo, Japan, Hong Kong and Visakhapatnam, India in the fashion industry, to produce products for the American Market. I began volunteering my time to remedy water quality issues with people in Pennsylvania, where my family lives, and built a non-profit

organization to focus on this work full-time. I have traveled extensively in Pennsylvania and New York to organize with communities, scientists, medical doctors and elected officials to change laws that are lenient toward the fossil fuel industry that compromises human health, wildlife health and air and water quality detriment.

(ZIESCHE) I have a bachelors in Telecommunications from the University of Florida School of Journalism with a minor in International Development and Humanitarian Assistance. I am currently in post-production for my full length documentary directorial debut. For three and a half years I was employed as the Grassroots Coordinator for International WOW Company's award winning GASLAND films. In that time I traveled around the world to communities being impacted by fracking and fracked gas infrastructure with a focus on communities in Pennsylvania and New York and often organized events with leading experts on the health and climate impacts of shale gas development. Since 2017 I've been working as the Community Engagement Coordinator for Sane Energy Project. During that time I've continued to travel to communities in New York and Pennsylvania being impacted by fracked gas infrastructure and produced over a dozen short documentary films telling their stories.

Q: What is your purpose in submitting this testimony?

A: The purpose of the testimony is to address the Company's Second Supplemental Testimony as it relates to Case 19-G-0678 and recent developments regarding implementation of state climate law and research on radioactivity in gas.

Q: Does this testimony contain exhibit?

Yes.

Exhibit_Sane 1
Exhibit_Sane 2
Exhibit_Sane 3

II. GAS ISSUES

A. Introduction and Background

Q: What will you address relating to the Company's business activities?

A. We have reviewed the Company's Second Supplemental Testimony, and this testimony addresses six issues relating to the Company's business activities.

First, we oppose the Company's proposal to accelerate construction of the Metropolitan Reliability Infrastructure (MRI) project and oppose the Company's request to raise rates to finish the project.

Second, we oppose the Company's proposal to expand the use of portable compressed natural gas (CNG) and any new investments to extend the life of CNG and liquified natural gas (LNG) facilities.

Third, we oppose ratepayer dollars being spent on the Clove Lakes Transmission Loop Project.

Fourth, we oppose ratepayer dollars being spent on the Riverhead Transmission and Southeast Suffolk Phase 2 projects.

Fifth, despite a small reduction in cost we still oppose the Company's proposals to spend hundreds of millions of ratepayer dollars on gas capital investments to replace Leak Prone Pipeline (LPP).

Sixth, taking into account new groundbreaking reporting in Rolling Stone we request the Company reconsider their position not to use funds for radon testing to ensure the safety of the Company's service.

Q. What is the overarching context in which the Company's gas business activities must be evaluated?

A.Shale gas production in the Appalachia region has increased rapidly since 2012 meaning an increasing amount of the gas supply coming into New York is extracted from the Marcellus Shale using high volume hydraulic fracking. Nationally, by 2016, 69%[1] of all wells drilled in this country were fracked and that number has continued to rise. The proposed Williams NESE Pipeline that much of the Company's rate case proposal supports would be transporting fracked gas from Pennsylvania. And in her Direct Testimony[2] Elizabeth D. Arangio confirmed more of

[1] See EIA *Hydraulically fractured horizontal wells account for most new oil and natural gas wells* https://www.eia.gov/todayinenergy/detail.php?id=33972

[2] Direct Testimony of Elizabeth D. Arangio before the Public Service Commission, April 2019 (p. 20)

the Company's gas will be fracked gas saying "as supplies from the Marcellus shale region became abundant and readily accessible, the Companies did not renew expiring long-haul contracts with Union, TransCanada, and Empire pipelines that delivered more expensive supplies from Dawn, Canada." Because of the extreme method used to develop gas from the Marcellus Shale, fracked gas has greater environmental impacts for the climate and on human health[3] than conventional gas and must be evaluated differently.

B. Metropolitan Reliability Infrastructure

Q.What did the Company propose regarding the MRI project in their Second Supplemental Testimony?

A: In their Second Supplemental Testimony the Company asserts "KEDNY's MRI Reliability project must be accelerated by one year to provide the hydraulic system support that is critical for enabling new customer connections over the next few winters".[4] The Company is proposing to spend an additional 184,670,417 ratepayer dollars to finish the pipeline.

Q.Do you support this proposal? Why or why not?

A: No we do not support this proposal because any investment to expand or extend the life of fracked gas infrastructure goes against climate science as well as city and state climate law to dramatically reduce greenhouse gas emissions. In response to SANE IR-4 the Company says "An assessment of greenhouse gas emissions was not done for this project" [5] and when asked if upstream methane emissions were taken into account the Company said "National Grid cannot control or measure upstream emissions on facilities it does not own and/or operate" yet the Climate Leadership and Community Protection Act (CLCPA) which Governor Andrew Cuomo signed into law on July 18, 2019 requires State agencies to take into account the full lifecycle of methane emissions which the Company confirms they are not doing.

As Sane Energy highlighted in our Direct Testimony "Cornell scientist Robert Howarth estimates that when looking at upstream and downstream emissions from shale gas, leakage rates range between 3.8% and 12%[6]. Even if upstream emissions were not taken into account Howarth

[3] The *Compendium of Scientific, Medical, and Media Findings Demonstrating Risks and Harms of Fracking (*the Compendium*)* is a fully referenced compilation of the evidence outlining the risks and harms of fracking. https://concernedhealthny.org/compendium/

[4] Second Supplemental Testimony of Gas Infrastructure and Operations Panel

[5] Exhibit_Sane 1

[6] See Robert W Howarth *Methane emissions and climatic warming risk from hydraulic fracturing and shale gas development: implications for policy*

estimates the downstream emissions associated with storing gas and delivering shale gas to market to be 2.5%. With leakage rates like this it is impossible to meet city and state goals to reduce greenhouse gas emissions and is more likely taking us in the wrong direction by increasing emissions." [7]

Q: Have there been any developments since August that make Howarth's research more relevant to this rate case? If so, what?

A: Yes, Robert Howarth has been named to the Climate Action Council which will advise on the implementation of the CLCPA.

Q. Have local elected officials made statements on the MRI project? If so, have they been in support or opposition?

Yes, State Senator Julia Salazar, Assemblymember Joe Lentol, City Council Member Antonio Reynoso, City Council Member Rafael Espinal, all local elected officials who have the pipeline within their district, as well as citywide elected officials Comptroller Scott Stringer and Public Advocate Jumaane Williams have made statements in opposition to the pipeline.

Q. Have community members made statements on the MRI project? If so, have they been in support or opposition?

As reported in the BKLYNR, Brooklyn Daily Eagle and Brooklyn Paper community members expressed opposition to the pipeline on January 14 at a Brooklyn Community Board -1 meeting. Community members specifically expressed safety and climate concerns as well as the Company's failure to do outreach on the project as reasons for opposition[8]. BKLYNR reported that "audience members were altogether dissatisfied with the responses from National Grid....They wanted to know why they weren't being heard, and why National Grid refused to give the community a platform in the decision process for this kind of project."[9] and Brooklyn

https://www.eeb.cornell.edu/howarth/publications/f_EECT-61539-perspectives-on-air-emissions-of-methane-and-climatic-warmin_100815_27470.pdf

[7] Sane Energy Direct Testimony
[8] Exhibit_Sane 2
[9] 'Public Utilities Under Public Control!' — North Brooklyn National Grid Pipeline Not Popular With Locals https://bklyner.com/north-brooklyn-pipeline/

Paper reported "How can you say you communicated anything to this community," said Antonia Ortiz. "No one knew about this until they had to get their car because they were digging a hole." [10]

Q:Has the public beyond the local community commented on the MRI project? And if so what have they said?

A: Yes, as of January 29, 2020 1,827 comments have been submitted to the PSC and a majority oppose the expansion of fracked gas infrastructure. Additionally over 3,800 New Yorkers have signed a petition[11] to Governor Cuomo and Commissioner John Rhodes calling on them to reject the Company's proposal to spend almost 185 million ratepayer dollars to complete the MRI project.

Q: What do you recommend?
We recommend that the public and local elected officials' call for an immediate halt to construction be listened to and no additional funds should be approved for the MRI project.

C. LNG & CNG

Q:What is the company proposing regarding CNG?

A: In their Second Supplemental Testimony the Company claims "For the winter 2020/2021, to meet increased gas demand from new and existing customers, the Glenwood CNG facility will be expanded to double its capacity. Two additional CNG injection facilities (a third and a fourth CNG facility) will be required in KEDLI's service territory at locations to be determined. At this point, five potential properties that are owned by National Grid in KEDLI's service territory are being considered. The capital costs for each new facility are estimated at approximately $1 million in FY 2020 and $10 million in FY 2021….Finally, the Companies require a CNG Filling Station to be located on Staten Island where trucks will be filled with CNG for transportation to both KEDLI's and KEDNY's CNG injection sites. The total cost for the project is approximately $30 million, and the project is expected to be in service in March of 2022." [12]

Q:Do you support this investment? Why or why not?

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[10] RESIDENTS FUME AS NAT GRID BUILDS NORTH BROOKLYN PIPELINE https://www.brooklynpaper.com/residents-fume-as-nat-grid-builds-north-brooklyn-pipeline/

[11] https://actionnetwork.org/petitions/tell-governor-cuomopublic-service-commission-listen-to-the-clcpa-reject-all-of-national-grids-fracked-gas-pipelines-and-proposals?source=direct_link&

[12] Second Supplemental Testimony of Gas Infrastructure and Operations Panel

A: No, we do not support the investments to expand CNG or build new CNG facilities because as stated in above testimony investments in fracked gas infrastructure go against climate science and city and state climate law. We recommend the Company invest more in gas demand reduction and energy efficiency.

Q:What is the company proposing regarding LNG?

In their Second Supplemental Testimony the Company claims "interim solutions to the capacity constraints involve additional reliance on the Companies' LNG capabilities, including updates to the Companies' existing LNG Tank Upgrade programs and to KEDNY's LNG Truck Load/Unload Station Program. Additionally, new investments are required for KEDNY's Greenpoint LNG site to add: (i) a new portable LNG facility and (ii) two additional vaporizers."[13]

Q:Do you support these investments? Why or why not?
No, we don't support these investments. While no investments in fracked gas infrastructure comply with city and state climate law, studies have found LNG increases emissions over regular gas because it is more intensive than the supply chain for pipeline gas, because of the extra processing steps such as processing, refrigeration, transportation, storage and handling. In order to meet our emission reduction benchmarks, we should be retiring LNG not investing in it.

Q: In addition to climate impacts are there other reasons the Company should plan on closing LNG & CNG facilities instead of investing in them?

A: Dangerous to surrounding communities and workers. LNG is highly volatile and explosive, and notably New York City saw a disaster that killed 40 workers[14].

Q: What do you recommend?

A: The Company should not invest any new funds that would expand the use of or extend the life of LNG facilities and should instead invest in gas demand reduction and energy efficiency.

D. Clove Lakes Transmission Loop Project

Q:What is the company proposing?

[13] Second Supplemental Testimony of Gas Infrastructure and Operations Panel
[14] https://www.silive.com/news/2013/02/40_years_ago_today_staten_isla.html

A: In their Second Supplemental Testimony the Company asserts "the Clove Lake Transmission Loop Project has been added to the FY 2021 and FY 2022 capital plans to enable KEDNY to conduct a feasibility study and engineering analyses. Looping of the Clove Lakes Transmission Main with a new 30-inch transmission main across the borough of Staten Island will facilitate KEDNY's ability to take more gas through the TETCO Goethals Take Station and ultimately move this gas across KEDNY's system to supply constrained areas." [15]

Q:Do you support this investment? Why or why not?

A: No we do not support the Company's proposal to invest 2 million ratepayer dollars to conduct a feasibility study for a pipeline expansion. As stated in the above testimony, investments in fracked gas infrastructure go against climate science and city and state climate law. The Company asserts the project "is not expected to be in service until FY 2025, at the earliest." In order to meet emission reduction goals of the CLCPA and Climate Mobilization Act New York will be reducing gas use leaving a pipeline coming into service in 2025 a stranded asset.

E. Riverhead Transmission and Southeast Suffolk projects

Q:What is the company proposing?

A: In it Second Supplemental Testimony the Company asserts "KEDLI's updates include adding back the Riverhead Transmission and Southeast Suffolk Phase 2 projects that were removed in the rebuttal testimony. These projects are required to support new customer connections in eastern Long Island. Moreover, to enable new customer connections, KEDLI has accelerated Phase 1 of the Southeast Suffolk project to ensure it will be in service in FY 2022. Accordingly, Exhibit __ (GIOP-1SS) reflects a shift of $5 million of the project forecast from FY 2022 to FY 2021." [16]

Q: What is the Riverhead Transmission project?

A: According to Direct Testimony filed by the Company in April "this project will install 8200 feet of 12" 350 psig transmission main on Mill Rd from County Rd 58 to Middle Rd, then on Middle Rd to Joise Ave." [17]

[15] Second Supplemental Testimony of Gas Infrastructure and Operations Panel
[16] Second Supplemental Testimony of Gas Infrastructure and Operations Panel
[17] Gas Infrastructure and Operations Panel

Q:Do you support this investment? Why or why not?

A: No we do not support the Company's proposal to invest over 3.5 million ratepayer dollars to install new transmission main. As stated in the above testimony, investments in fracked gas infrastructure go against climate science and city and state climate law. The Company asserts the project "will be constructed in FY 2024."[18] In order to meet emission reduction goals of the CLCPA and Climate Mobilization Act New York will be reducing gas use leaving a pipeline being constructed in 2024 a stranded asset.

Q: What is the Southeast Suffolk project?

A:According to Direct Testimony filed by the Company in April the project will "install approximately 8,500 feet of 16-inch, 350 psig transmission main, a new regulator station, and 5,800 feet of 12-inch steel distribution main. The Company's capital investment plan includes approximately $20 million in the Rate Year, $21.6 million in Data Year 1, $0.1 million in Data Year 2, and $0.3 million in Data Year 3."[19]

Q:Do you support this investment? Why or why not?

A:No we do not support the Company's proposal to invest over 42.2 million ratepayer dollars to install a new transmission main, a new regulator station and new distribution main. As stated in the above testimony, investments in fracked gas infrastructure go against climate science and city and state climate law.

F. Leak Prone Pipe

Q:What was the Company's update regarding LPP?

A:In it Second Supplemental Testimony the Company asserts "KEDLI renegotiated its agreement with a contractor that provides services for the Proactive Main Replacement (LPP) Program. A copy of the renegotiated agreement was provided in the Company's response to IR No. DPS-1043.The agreement results in reduction of the program's unit costs from $306 per foot to $304 per foot, as shown in Exhibit (GIOP-4SS). Revising the cost per foot reduces KEDLI's capital expense forecast by $1.929 million in the Rate Year, as shown in Exhibit (GIOP-1SS)."[20]

[18] Second Supplemental Testimony of Gas Infrastructure and Operations Panel
[19] Gas Infrastructure and Operations Panel
[20] Second Supplemental Testimony of Gas Infrastructure and Operations Panel

Q: Does the reduction in cost change your position on LPP?

A:No. We still argue that the Company should not invest any new ratepayer funds in extending the life of pipelines.The Company should be planning on retiring LPP instead of replacing it.

G: Need for Radon Testing

Q: What was Sane Energy's recommendation on radon testing in testimony filed in August?

In Direct Testimony Sane asserted "In order for the Company to comply with its requirement to provide safe service we recommend the Company provide EPA scientific radon monitoring[21] in all gas supplied to customers. Additionally, we recommend meeting with New York elected officials who drafted the 2014 Radon Bill[22] to monitor and mitigate radon in our gas supply." [23]

Q:Why did Sane make that recommendation?

A: We made that recommendation because the Company asserted in Direct Testimony[24] that more of their gas supply is coming from the Marcellus Shale in northeastern Pennsylvania which has higher than average levels of uranium and thorium. These elements break down to form radium-226. Signs of radon in gas & other fracked gas chemicals in our homes, have been on the rise since 2004[25]when high-volume hydraulic fractured drilling began. Radon is the leading cause of lung cancer in non-smokers in the United States[26].

Q: Have there been any significant developments regarding radioactivity in gas since Sane Energy Project filed testimony in August? If so, what are the developments?

A: Yes. A groundbreaking, 20-month investigation on oil and gas radioactivity by reporter Justin Nobel was published in Rolling Stone on January 21, 2020[27].

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[21] https://www.epa.gov/radon
[22] https://nyassembly.gov/leg/?default_fld=&bn=A06863&term=2013&Summary=Y&Text=Y
[23] Sane Direct Testimony
[24] Direct Testimony of Elizabeth D. Arangio before the Public Service Commission, April 2019 (p. 20)
[25] Increased Levels of Radon in Pennsylvania Homes Correspond to Onset of Fracking Johns Hopkins Bloomberg School of Public Health
https://www.jhsph.edu/news/news-releases/2015/increased-levels-of-radon-in-pennsylvania-homes-correspond-to-onset-of-fracking.html
[26] https://www.cancer.org/cancer/cancer-causes/radiation-exposure/radon.html
[27] America's Radioactive Secret
https://www.rollingstone.com/politics/politics-features/oil-gas-fracking-radioactive-investigation-937389/

Q:What did the investigation find?

A: Nobel's reporting found that major fossil fuel companies have known for decades that gas has high levels of naturally occurring radioactivity and that radioactivity builds up within their infrastructure. Nobel wrote "In the early 1970s, Exxon learned radioactivity was building up in pumps and compressors at most of its gas plants. "Almost all materials of interest and use to the petroleum industry contain measurable quantities of radionuclides," states a never-publicly released 1982 report by the American Petroleum Institute, the industry's principal trade group, passed to *Rolling Stone* by a former state regulator...Despite dire climate warnings, the U.S. oil-and-gas industry is in the midst of an epic boom, what a 2018 Department of Energy paper calls an "oil-and-gas production renaissance." Pipelines, power plants, and shipping terminals are being developed across the nation at a dizzying pace. But in the excitement of this boom there is little mention of the pipes, pumps, and filters in these plants that will become coated with radioactivity."

Q: Is the Company testing for radon?

A: No. In response to Sane IR-3 when asked if testing for radon the Company said "where gas is introduced into the National Grid system, gas monitoring instruments are needed to monitor odorant levels, BTU, composition, hydrates, and hydrocarbon dew point ("HCDP")." and went further denying the radioactivity of the Marcellus Shale saying "the Companies do not accept the conclusion asserted in the question that "Marcellus Shale has been proven to have high levels of radon and hydraulic fracking uses dozens of carcinogenic chemicals". [28]

Q: What do you recommend?

A: We recommend the Company do radon testing in accordance with bill A1020[29] which has been introduced in the New York State Assembly.

[28] Exhibit_Sane 3
[29] https://www.nysenate.gov/legislation/bills/2019/A1020